

Two Scoreboards - Cleaning BEADs Label Noise with QLoRA on Llama-3.1-8B

IMT 526 - Spring 2026

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Goal: Binary bias classifier for news text

BEADs

The most
comprehensive public
dataset

Method

QLoRA-tuned
Llama-3.1-8B,
swept training sizes

Baseline

0.804 accuracy (beat
BEAD paper's
Llama-2 by ~3 pts)

What we discovered

First crack: Nobody transfers

- Diagonal mean 0.842 - strong on own data
- Off-diagonal mean 0.510 - a coin flip
- First read: "four datasets measure four different things"

Train \ Eval	BEADs	BABE	cajcodes	WNC
BEADs	0.799	0.312	0.667	0.462
BABE	0.379	0.867	0.424	0.563
cajcodes	0.422	0.588	0.970	0.557
WNC	0.407	0.760	0.576	0.734

We built our own ground truth

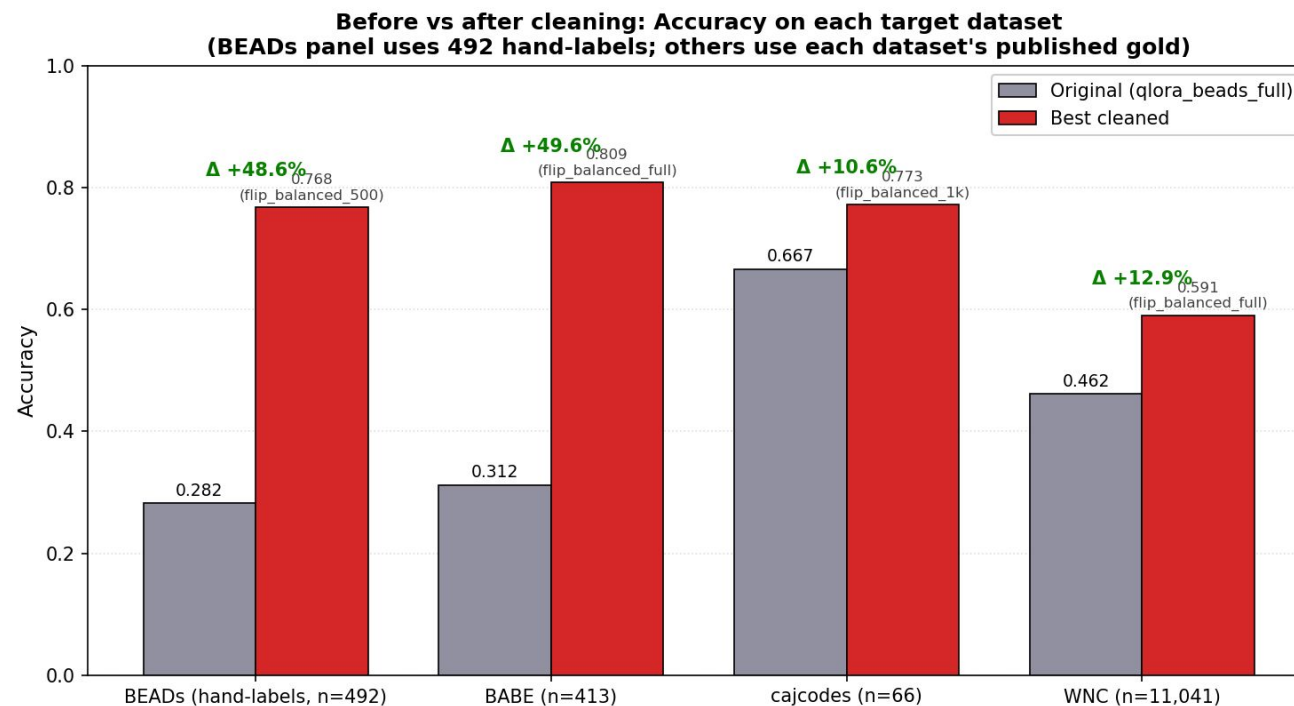
- Hand-labeled 492 rows, blind to gold and to the model
- Gate first: one calibration round, then IAA gate passed; κ 0.57 - 0.75
- Example: A 2nd-Amendment rant
- Labeled **unbiased** by BEADs

Example

"This is NOT FUNNY! This is the 2nd Amendment! YOU don't want machine gun turrets on cars, then YOU go ahead and live in your little libtard Hillary/Bernie loving threesome hiding emails. You don't OWN the 2nd Amendment any more than you OWN the Social Security Trust Fund..."

The diagnosis: Same model, 0.80 vs 0.28

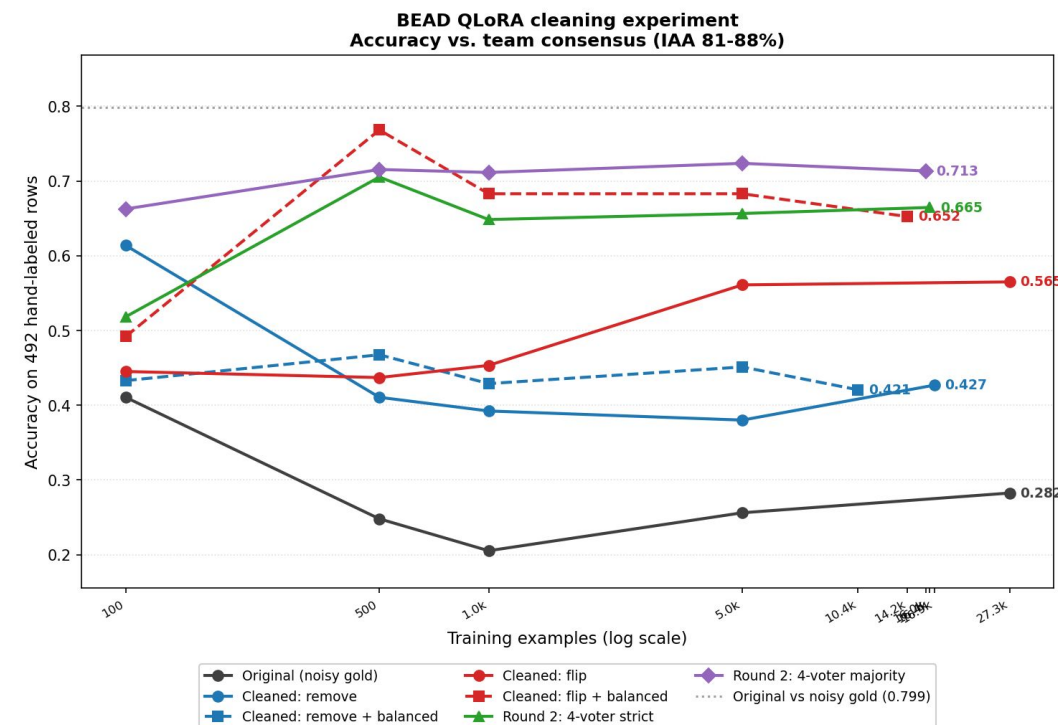
- BEADs gold disagrees with our consensus on 69.5% of rows
- Same model: ~0.80 vs BEADs / 0.28 vs humans (below chance)
- BEADs over-calls bias ~1.3×



How we fixed it

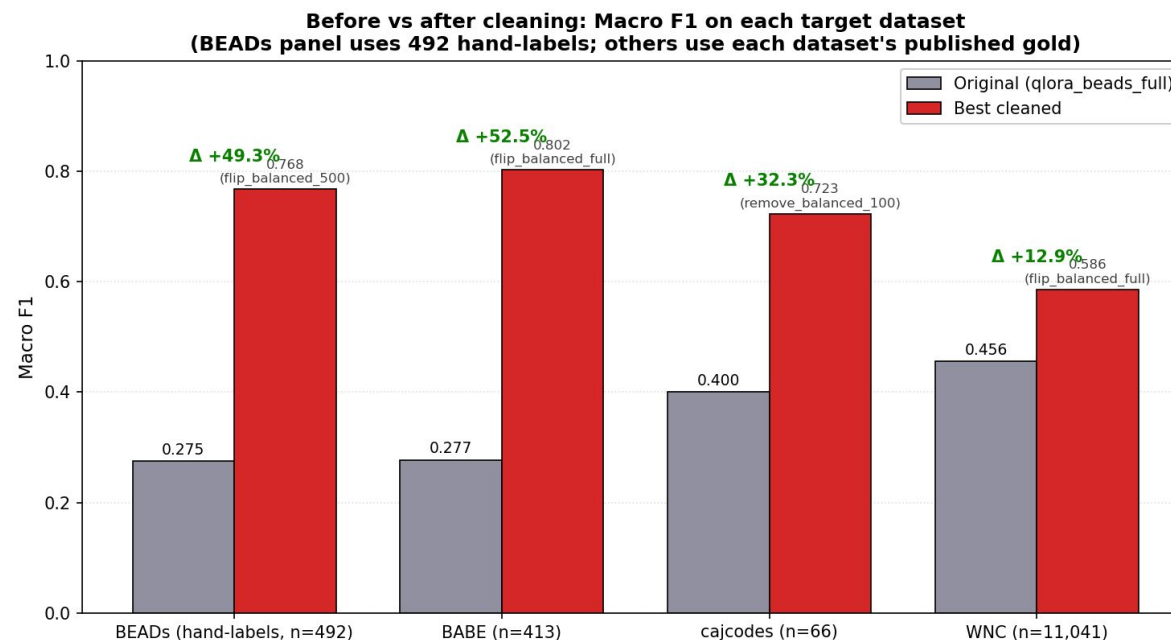
The fix: Let outside datasets vote

- Rule: 3 external adapters (BABE, cajcodes, WNC) unanimously disagree with gold → flag + relabel to their vote
- Flagged 38% of training rows (10,371 / 27,263)
- Flips matched human truth 85%
- Rebalanced classes 50/50, retrained



The payoff: +49 points, and it transfers

- vs human labels: 0.28 → stable 0.72–0.77 band (+44 - 49 pp)
- 0.768 peak = partly sampling variance; 0.724 = stable ceiling
- Transfer to BABE (never trained on): 0.312 → 0.809 (specialist 0.867)



What it means, and what we still owe

- Takeaway: Scoring on a dataset's own labels overstates real performance on subjective tasks
- Limits: 492 labels ($\sim\pm 4$ pp CI); $\sim 15\%$ of relabels still wrong; "public-interest bias" is one definition
- Next: Larger validation set, confidence-weighted relabeling, active learning

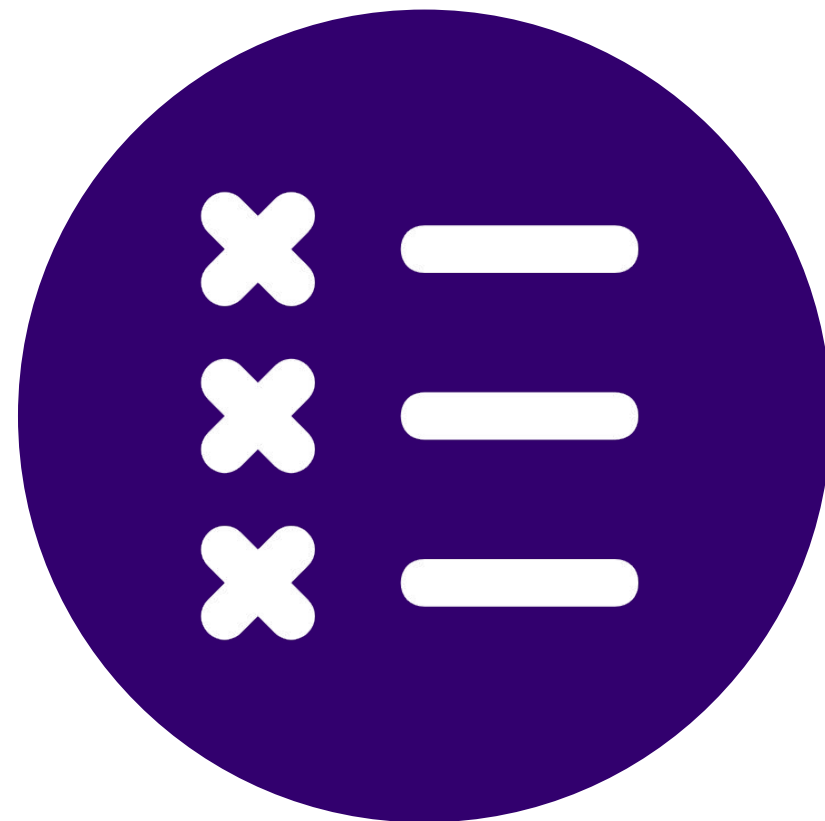
Thank you!

References

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- Zhang, R., Li, Y., Ma, Y., Zhou, M., and Zou, L. (2023). LLMaAA: Making large language models as active annotators. *arXiv:2310.19596*.

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- **Dataset:** BEADs — most comprehensive public bias dataset
- **Method:** QLoRA-tuned Llama-3.1-8B, swept training sizes
- **Baseline:** 0.804 accuracy (beat BEAD paper's Llama-2 by ~3 pts)

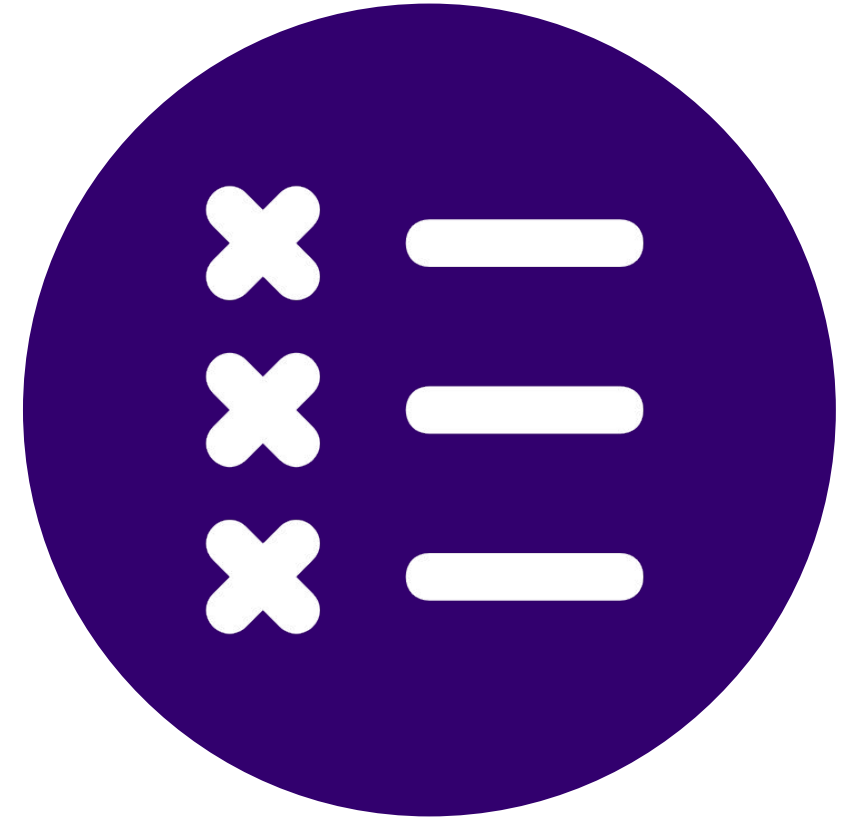


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Label issues - Abrevaa - 30 secs

- BEADs labels incorrectly
- **Example:** "This is NOT FUNNY! This is the 2nd Amendment! YOU don't want machine gun turrets on cars, then YOU go ahead and live in your little libtard Hillary/Bernie loving threesome hiding emails. You don't OWN the 2nd Amendment any more than you OWN the Social Security Trust Fund..."
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